

$$\begin{aligned} & \mathbb{P} \\ & Z_i = \\ & I(X_i \geq \\ & x_0) \\ & X_i \\ & x_0 \\ & \{Y_i(1), Y_i(0)\} \\ & \tau_Y(x_0) = \lim_{\varepsilon \rightarrow 0+} E(Y \mid X = x_0 + \varepsilon) - \lim_{\varepsilon \rightarrow 0+} E(Y \mid X = x_0 - \varepsilon). \end{aligned}$$

(1)

$$\begin{aligned} & Z_i = \\ & I(X_i \geq \\ & x_0) \\ & Z_i \\ & D_i \\ & X_i \geq \\ & x_0 \\ & Z_i \\ & D_i \\ & Y_i \\ & X_i \\ & x_0 \end{aligned}$$

$$Z_i = I(X_i \geq x_0).$$

$$\begin{aligned} & D_i \\ & Z_i \\ & P(D_i = \\ & 1 \mid \\ & X_i = \\ & x) \\ & x_0 \\ & Y_i \\ & Z_i \\ & \{D_i(1), D_i(0), Y_i(1), Y_i(0)\} \\ & \tau_D(x_0) = E\{D(1) - D(0) \mid X = x_0\} \end{aligned}$$

$$\tau_Y(x_0) = E\{Y(1) - Y(0) \mid X = x_0\}.$$

(2)

$$\begin{aligned} & Z \\ & D \\ & X = \\ & x_0 \\ & Z \\ & I(\bar{X} \geq \\ & x_0) \\ & x_0 \end{aligned}$$

$$D_i(1) \geq D_i(0),$$

(3)

$$D_i(1) = D_i(0) \implies Y_i(1) = Y_i(0),$$

(4)

$$\begin{aligned} & x_0 \\ & E\{D(1) \mid \\ & X = \\ & x\} \\ & E\{Y(1) \mid \\ & X = \\ & x\} \\ & x_0 \\ & E\{D(0) \mid \\ & X = \\ & x\} \\ & E\{Y(0) \mid \\ & X = \\ & x\} \\ & x_0 \end{aligned}$$

$$\tau_{\text{LATE}}(x_0) = E\{Y(1) - Y(0) \mid D(1) > D(0), X = x_0\},$$

(5)

$$\tau_{\text{LATE}}(x_0) = \frac{E\{Y(1) - Y(0) \mid X = x_0\}}{E\{D(1) - D(0) \mid X = x_0\}}.$$

(6)

$$\begin{aligned} & E(D \mid \\ & X = \\ & x) \\ & x_0 \\ & \tau_{\text{LATE}}(x_0) = \frac{\lim_{\varepsilon \rightarrow 0+} E(Y \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \rightarrow 0+} E(Y \mid Z = 0, X = x_0 - \varepsilon)}{\lim_{\varepsilon \rightarrow 0+} E(D \mid Z = 1, X = x_0 + \varepsilon) - \lim_{\varepsilon \rightarrow 0+} E(D \mid Z = 0, X = x_0 - \varepsilon)}. \end{aligned}$$

(7)

$$\begin{aligned} & \text{th:frdd-} \\ & \text{identification} \\ & Z \\ & h \\ & x_0 \\ & X_i \in \end{aligned}$$